ECON 101: Macroeconomics
Solutions – Quiz 3

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October 22, 2025

Part A – Multiple Choice Solutions (1 mark each)

1. CPI inflation from 152 to 160
Inflation $= \dfrac{160-152}{152} \times 100 = \dfrac{8}{152} \times 100 \approx 5.26\% \Rightarrow 5.3\%$.
Answer: (B).

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2. Fisher approximation
Real interest $\approx i - \pi = 5\% - 3\% = 2\%$. (Exact Fisher: $\dfrac{1.05}{1.03} - 1 \approx 1.94\%$.)
Answer: (B).

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3. CPI coverage
The CPI covers the prices of goods and services purchased by consumers, including imports, utilities, restaurant meals, *and* used vehicles. With the options given, all listed items are typically included in the CPI.
Grading note: Since no "None of the above" option was provided, accept that **all options are included in CPI**. If a single key is required, flag this item for revision (replace with a clearly excluded item such as "corporate bond purchases").

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4. Catch-up time: growth differential
Solve for t in $28{,}000(1.04)^t = 42{,}000(1.02)^t$.
$\left(\frac{1.04}{1.02}\right)^t = \frac{42{,}000}{28{,}000} = 1.5.$
$t = \frac{\ln(1.5)}{\ln(1.04/1.02)} \approx \frac{0.405465}{0.019414} \approx 20.9 \text{ years} \approx 21 \text{ years}.$
Answer: (A).

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5. Early mover in recessions
Investment is the most volatile component of spending and typically falls first.
Answer: (D).

Part B – Long Problem Solutions (15 marks)

6. Economic Growth and Inflation Analysis

Data:

Item	Qty (Y1)	Qty (Y2)	Price (Y1)	Price (Y2)
Smartphones	200	250	\$500	\$520
Tablets	100	120	\$800	\$880
Headphones	300	330	\$100	\$90

(a) Nominal GDP, Real GDP (Y2 in Y1 prices), and Real Growth

Step 1: Nominal GDP (current-year prices)
Year 1:

$$\text{NGDP}_1 = (200)(500) + (100)(800) + (300)(100) = 100{,}000 + 80{,}000 + 30{,}000 = \boxed{210{,}000}$$

Year 2:

$$\text{NGDP}_2 = (250)(520) + (120)(880) + (330)(90) = 130{,}000 + 105{,}600 + 29{,}700 = \boxed{265{,}300}$$

Step 2: Real GDP for Year 2 in Year 1 prices (base Y1)
Hold prices at Y1 levels, update quantities to Y2:

$$\text{RGDP}_{2|Y1 \text{ prices}} = (250)(500)+(120)(800)+(330)(100) = 125{,}000+96{,}000+33{,}000 = \boxed{254{,}000}$$

Note: RGDP_1 at Y1 prices equals $\text{NGDP}_1 = \boxed{210{,}000}$.

Step 3: Real GDP growth (base Y1)

$$\text{Real growth} = \frac{254{,}000 - 210{,}000}{210{,}000} \times 100 = \frac{44{,}000}{210{,}000} \times 100 \approx \boxed{20.95\% \ (\approx 21\%)}$$

(b) GDP Deflator and Inflation

Step 1: GDP Deflator

$$\text{Deflator}_t = \frac{\text{NGDP}_t}{\text{RGDP}_t} \times 100$$

Year 1: $\text{Deflator}_1 = \dfrac{210{,}000}{210{,}000} \times 100 = \boxed{100.00}$.

Year 2: $\text{Deflator}_2 = \dfrac{265{,}300}{254{,}000} \times 100 \approx \boxed{104.45}$.

Step 2: Inflation (deflator-based)

$$\pi \approx \frac{104.45 - 100.00}{100.00} \times 100 = \boxed{4.45\%}$$

(c) Real GDP per Capita Growth

Population: $N_1 = 800$, $N_2 = 820$.

$$\text{RGDPpc}_1 = \frac{210{,}000}{800} = \boxed{262.50} \quad \text{and} \quad \text{RGDPpc}_2 = \frac{254{,}000}{820} \approx \boxed{309.76}$$

$$\text{Growth in RGDP per capita} = \frac{309.76 - 262.50}{262.50} \times 100 \approx \boxed{18.10\%}$$

(d) Living Standards Assessment

Real GDP grew by about 21%, population rose by 2.5% (from 800 to 820), and *real GDP per capita* increased by about 18.1%. Despite moderate inflation (about 4.45% via the deflator), the substantial rise in *real* output per person indicates that living standards **improved markedly**.

End of Instructor Solutions